Graviton Field $H(r,\theta,t)$ in the TauTimp Framework Spherical wave equation with variable coefficient R/r: separation of variables, Airy-function eigenmodes, and construction of H 20Mai2026

Framework anchors. - Onduscular radius: $R = 4.2355 \times 10^{22}$ m. - Inner cutoff (space quantum): $r_0 = 4.3 \times 10^{-19}$ m. - Speed of light:c - Reduced Planck constant: $\hbar = 1.054571817 \times 10^{-34}$ J s.

Note on the inner anchor. In earlier TauTimp work the inner cutoff was identified with the Schwarzschild radius $r_0 \approx 1.12 \times 10^4$ m. Here the inner BC is shifted to the **space-quantum scale** $r_0 = 4.3 \times 10^{-19}$ m, representing the smallest meaningful spatial discretization of the onduscular medium. This is 23 orders of magnitude finer than the Schwarzschild anchor and ~ 16 orders coarser than the Planck length.

1. PDE structure under spherical symmetry

The variable-coefficient wave equation is $\frac{R}{r}\,\Delta H(r,\theta,t) = \frac{1}{c^2}\frac{\partial^2 H}{\partial t^2}$.

Under full spherical symmetry H depends on (r,t) only; we therefore solve for $f(r,t)$ defined by $H(r,\theta,t) \equiv f(r,t)$.

The spherically-symmetric Laplacian is $\Delta f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) = \frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\frac{\partial f}{\partial r}$.

Multiplying by R/r yields the governing equation $\boxed{\frac{R}{r}\frac{\partial^2 f}{\partial r^2} + \frac{2R}{r^2}\frac{\partial f}{\partial r} = \frac{1}{c^2}\frac{\partial^2 f}{\partial t^2}.}$

Boundary conditions: $f(r_0,t) = 0, \qquad f(2R,t) = 0.$

2. Separation of variables

Set $f(r,t) = g(r)\,u(t)$. Substituting into (1.3) and dividing by f:

$$\frac{R}{r}\frac{g''(r)}{g(r)} + \frac{2R}{r^2}\frac{g'(r)}{g(r)} = \frac{1}{c^2}\frac{u''(t)}{u(t)} = -k^2c^2\cdot\frac{1}{c^2} = -k^2.$$

That is, we introduce the separation constant $-k^2$ (with k real for oscillatory temporal modes).

2.1 Temporal factor

$$u''(t) + k^2c^2\,u(t) = 0 \;\Rightarrow\; u(t) = C_1\cos(\omega t) + C_2\sin(\omega t), \quad \omega \equiv kc.$$

2.2 Radial equation

$$\frac{R}{r}\,g''(r) + \frac{2R}{r^2}\,g'(r) + k^2\,g(r) = 0.$$

Multiplying (2.3) by r/R: $g''(r) + \frac{2}{r}\,g'(r) + \frac{k^2}{R}\,r\,g(r) = 0.$

3. Reduction to Airy's equation

Apply the substitution $g(r) = h(r)/r$, so $g' = \frac{h'}{r} - \frac{h}{r^2}, \qquad g'' = \frac{h''}{r} - \frac{2h'}{r^2} + \frac{2h}{r^3}$.

Substituting into (2.4): $\left(\frac{h''}{r} - \frac{2h'}{r^2} + \frac{2h}{r^3}\right) + \frac{2}{r}\left(\frac{h'}{r} - \frac{h}{r^2}\right) + \frac{k^2 r}{R}\cdot\frac{h}{r} = 0,$

which simplifies (the h' and $1/r^3$ pieces cancel exactly) to $\boxed{h''(r) + \frac{k^2 r}{R}\,h(r) = 0.}$

This is the **Airy equation in disguise**. Introducing

$$\alpha \equiv \left(\frac{k^2}{R}\right)^{1/3}, \qquad \xi \equiv -\alpha\,r,$$

we have $d/dr = -\alpha\,d/d\xi$, so $h'' = \alpha^2\,d^2h/d\xi^2$, and (3.1) becomes

$$\alpha^2\frac{d^2h}{d\xi^2} + \frac{k^2}{R}\left(-\frac{\xi}{\alpha}\right)h = 0 \;\Rightarrow\; \frac{d^2h}{d\xi^2} = \xi\,h,$$

the standard Airy equation. Its general solution is

$$h(r) = c_1\,\mathrm{Ai}(\xi(r)) + c_2\,\mathrm{Bi}(\xi(r)), \qquad \xi(r) = -\alpha\,r.$$

Hence $\boxed{g(r) = \frac{1}{r}[\,c_1\,\mathrm{Ai}(-\alpha r) + c_2\,\mathrm{Bi}(-\alpha r)].}$

4. Eigenvalue problem

The two boundary conditions $g(r_0) = 0$ and $g(2R) = 0$, equivalently $h(r_0) = 0$ and $h(2R) = 0$, give the linear system $\begin{pmatrix} \mathrm{Ai}(-\alpha r_0) & \mathrm{Bi}(-\alpha r_0) \\ \mathrm{Ai}(-\alpha\,2R) & \mathrm{Bi}(-\alpha\,2R) \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = 0.$

Non-trivial solutions require $\boxed{D(k) \equiv \mathrm{Ai}(-\alpha r_0)\,\mathrm{Bi}(-\alpha\,2R) - \mathrm{Ai}(-\alpha\,2R)\,\mathrm{Bi}(-\alpha r_0) = 0.}$

This transcendental equation determines the discrete eigenvalues k_n, hence the eigenfrequencies $\omega_n = k_n c$.

4.1 Behavior at r_0

Because $r_0/R \sim 10^{-41}$, the argument $-\alpha r_0$ is extremely close to zero for all eigenvalues of practical interest. Using

$$\mathrm{Ai}(0) = \frac{1}{3^{2/3}\,\Gamma(2/3)} \approx 0.3550, \qquad \mathrm{Bi}(0) = \frac{1}{3^{1/6}\,\Gamma(2/3)} \approx 0.6149,$$

the inner-BC ratio satisfies $\frac{\mathrm{Ai}(-\alpha r_0)}{\mathrm{Bi}(-\alpha r_0)} \underset{r_0\to 0}{\to} \frac{1}{\sqrt{3}} \approx 0.57735.$

Hence to excellent approximation, the eigenvalue condition reduces to $\boxed{\frac{\mathrm{Ai}(-\alpha\,2R)}{\mathrm{Bi}(-\alpha\,2R)} = \frac{1}{\sqrt{3}}.}$

4.2 WKB asymptotic formula

For large negative argument, the Airy functions admit the asymptotic forms

$$\mathrm{Ai}(-z) \sim \pi^{-1/2} z^{-1/4} \sin\left(2/3\, z^{3/2} + \pi/4\right), \quad \mathrm{Bi}(-z) \sim \pi^{-1/2} z^{-1/4} \cos\left(2/3\, z^{3/2} + \pi/4\right),$$

valid for $z \to +\infty$. Inserting $z = \alpha r$ for both BCs into (4.2), the determinant reduces to

$$D(k) \propto \sin\left[2/3\, \alpha^{3/2} \left((2R)^{3/2} - r_0^{3/2}\right)\right] = 0,$$

so eigenvalues satisfy $2/3\, \alpha^{3/2} \left((2R)^{3/2} - r_0^{3/2}\right) = n\pi, \qquad n = 1,2,3, \ldots$

Using $\alpha^{3/2} = k/\sqrt{R}$ and dropping $r_0^{3/2}$ (since $r_0 \ll R$): $\boxed{k_n^{\mathrm{WKB}} = \frac{3n\pi}{2\,(2R)^{3/2}} \sqrt{R} = \frac{3\,n\,\pi}{4\sqrt{2}\,R}.}$

Numerically $k_1^{\mathrm{WKB}} = 3.934 \times 10^{-23}\ \mathrm{m}^{-1}$.

4.3 Numerical eigenvalues

Solving (4.2) exactly via fsolve/brentq on the determinant function $D(k)$ gives:

n	k_n [m^{-1}]	$\omega_n = k_n c$ [rad s^{-1}]	$T_n = 2\pi/\omega_n$ [s]	T_n [yr]	k_n/k_n^{WKB}
1	3.6343×10^{-23}	1.0895×10^{-14}	5.7668×10^{14}	1.827×10^{7}	0.9239
2	7.5536×10^{-23}	2.2645×10^{-14}	2.7746×10^{14}	8.791×10^{6}	0.9601
3	1.1482×10^{-22}	3.4423×10^{-14}	1.8253×10^{14}	5.783×10^{6}	0.9730
4	1.5414×10^{-22}	4.6209×10^{-14}	1.3597×10^{14}	4.308×10^{6}	0.9796
5	1.9346×10^{-22}	5.7997×10^{-14}	1.0834×10^{14}	3.432×10^{6}	0.9836
6	2.3278×10^{-22}	6.9787×10^{-14}	9.0034×10^{13}	2.853×10^{6}	0.9863
7	2.7211×10^{-22}	8.1578×10^{-14}	7.7021×10^{13}	2.440×10^{6}	0.9882
8	3.1144×10^{-22}	9.3369×10^{-14}	6.7294×10^{13}	2.132×10^{6}	0.9897
9	3.5078×10^{-22}	1.0516×10^{-13}	5.9749×10^{13}	1.893×10^{6}	0.9908
10	3.9011×10^{-22}	1.1695×10^{-13}	5.3725×10^{13}	1.703×10^{6}	0.9917

The WKB formula (4.6) is accurate to better than 1% for $n \geq 10$ and within ~ 8% even for the ground mode.

4.4 Comparison with the d'Alembert timescale R/c

The fundamental period is $T_1 = 2\pi/(k_1 c) \approx 5.77 \times 10^{14}$ s $\approx 18.3\ Myr$, about 4π times the d'Alembert crossing time $R/c \approx 4.48\ Myr$; specifically $T_1/(R/c) \approx 4.08$ as expected from (4.6): $T_1/(R/c) = 2\pi/(k_1 R) = 8\sqrt{2}/3 \approx 3.77$ in the WKB limit.

5. Eigenfunctions

For each eigenvalue k_n, the eigenfunction is

$$h_n(r) = \mathcal{N}_n\left[\mathrm{Ai}(-\alpha_n r) - 1/\sqrt{3}\ \mathrm{Bi}(-\alpha_n r)\right],$$

where the inner-BC ratio (4.3) has been imposed (i.e., we picked $c_1 = \mathcal{N}_n$, $c_2 = -\mathcal{N}_n/\sqrt{3}$). The radial mode is then

$$g_n(r) = \frac{h_n(r)}{r}.$$

5.1 Normalization

Choosing the radial inner product $\langle g_m, g_n \rangle \equiv \int_{r_0}^{2R} g_m^*(r)\, g_n(r)\, r^2\, dr$ (the natural spherical L^2 inner product), and using $g_n = h_n/r$: $\int_{r_0}^{2R} |g_n(r)|^2\, r^2\, dr = \int_{r_0}^{2R} |h_n(r)|^2\, dr \overset{!}{=} 1$,

So $\boxed{\mathcal{N}_n = \left[\int_{r_0}^{2R} \left(\mathrm{Ai}(-\alpha_n r) - 1/\sqrt{3}\ \mathrm{Bi}(-\alpha_n r)\right)^2 dr\right]^{-1/2}.}$

For the ground mode ($n = 1$), numerical evaluation gives $\mathcal{N}_1 \approx 8.555 \times 10^{-12}$.

5.2 WKB asymptotic eigenfunction

In the bulk region ($\alpha_n r \gg 1$), the WKB form is

$$h_n^{\mathrm{WKB}}(r) \propto (\alpha_n r)^{-1/4} \sin\left[2/3\, (\alpha_n r)^{3/2} - 2/3\, (\alpha_n r_0)^{3/2}\right].$$

This reproduces the exact eigenfunction to good accuracy except in a narrow region near the turning point at $r = 2R$ (where the Airy oscillation ceases).

6. General solution and construction of $H(r,\theta,t)$

The general solution of (1.3) is the superposition $f(r,t) = \sum_{n=1}^{\infty} [A_n \cos(\omega_n t) + B_n \sin(\omega_n t)] g_n(r)$, with $\omega_n = k_n c$ and g_n from (5.2). The coefficients A_n, B_n are determined by the initial data $f(r,0)$ and $\partial_t f(r,0)$ via the orthogonality $\int_{r_0}^{2R} g_m(r)\, g_n(r)\, r^2\, dr = \delta_{mn}$.

6.1 Spherically-symmetric H

By assumption, $H(r,\theta,t) = f(r,t)$, i.e., independent of the polar angle θ. The full graviton field is therefore

$$\boxed{H(r,\theta,t) = \sum_{n=1}^{\infty} [A_n \cos(\omega_n t) + B_n \sin(\omega_n t)] \frac{1}{r} \left[\mathrm{Ai}(-\alpha_n r) - 1/\sqrt{3}\, \mathrm{Bi}(-\alpha_n r)\right] \mathcal{N}_n.}$$

6.2 Note on angular structure

Had we relaxed spherical symmetry, the PDE (1.1) would admit a full angular expansion

$$H(r,\theta,\varphi,t) = \sum_{n,\ell,m} c_{n\ell m}(t)\, g_{n\ell}(r)\, Y_\ell^m(\theta,\varphi),$$

where the radial functions $g_{n\ell}(r)$ would satisfy the ℓ-dependent modification of (2.4):

$$g''_{n\ell} + \frac{2}{r} g'_{n\ell} - \frac{\ell(\ell+1)}{r^2} g_{n\ell} + \frac{k_{n\ell}^2}{R} r\, g_{n\ell} = 0.$$

Under spherical symmetry only $\ell = 0$ contributes (with $Y_0^0 = 1/\sqrt{4\pi}$), and this case reduces exactly to (3.1). The present derivation is therefore the $\ell = 0$ mode of the general expansion.

7. Summary table

Quantity	Symbolic	Value (or method)
PDE	$(R/r)\Delta H = (1/c^2)\, H_{tt}$	—
Separation constant	$-k^2$	—
Reduced radial eq.	$h'' + (k^2 r/R)\, h = 0$	Airy form
Airy variable	$\xi = -\alpha r,\ \alpha = (k^2/R)^{1/3}$	—
Eigenvalue condition	$\mathrm{Ai}(-\alpha r_0)\, \mathrm{Bi}(-2\alpha R) - \mathrm{Ai}(-2\alpha R)\, \mathrm{Bi}(-\alpha r_0) = 0$	(4.2)
Inner ratio (since $r_0 \to 0$)	$\mathrm{Ai}(-\alpha r_0)/\mathrm{Bi}(-\alpha r_0) \to 1/\sqrt{3}$	(4.3)
Asymptotic eigenvalues	$k_n^{\mathrm{WKB}} = 3n\pi/(4\sqrt{2}\, R)$	(4.6)
Ground eigenvalue (numerical)	k_1	$3.634 \times 10^{-23}\ \mathrm{m}^{-1}$
Ground frequency	$\omega_1 = k_1 c$	$1.090 \times 10^{-14}\ \mathrm{rad\,s}^{-1}$
Ground period	$T_1 = 2\pi/\omega_1$	$5.767 \times 10^{14}\ \mathrm{s} \approx 18.3\ Myr$
Ground normalization	$\mathcal{N}_1$	8.555×10^{-12}
Radial mode	$g_n(r) = h_n(r)/r$	—
Eigenfunction	$h_n(r) = \mathcal{N}_n[\mathrm{Ai}(-\alpha_n r) - \mathrm{Bi}(-\alpha_n r)/\sqrt{3}]$	(5.1)
Full field	$H = \sum_n [A_n \cos\omega_n t + B_n \sin\omega_n t]\, g_n(r)$	(6.3)

8. Physical interpretation

1. **Variable coefficient** R/r in (1.1) introduces a position-dependent effective wave speed: regions near the origin propagate gravitational disturbances "faster" (in the sense of the multiplier $R/r > 1$ for $r < R$) than far regions. This is a signature of the onduscular medium's anisotropy in r.
2. **The reduction to Airy's equation (3.1)** is exact and shows that the radial graviton modes share a deep mathematical structure with the modified Newton problem analyzed in the earlier GravitonPaper work, supporting consistency of the TauTimp framework across both static-force and dynamical-wave perspectives.
3. **The inner cutoff** $r_0 = 4.3 \times 10^{-19}$ m has negligible impact on the eigenvalues (only the limiting ratio $1/\sqrt{3}$ from $\mathrm{Ai}(0)/\mathrm{Bi}(0)$ matters), but enforces regularity of the field on a discrete spatial lattice — a quantum-of-space interpretation.
4. **The cosmological boundary** $r = 2R$ sets the natural scale for the spectrum. The fundamental period $T_1 \approx 18\ Myr$ is well below the Hubble time (~ 8.2 Gyr) by a factor ~ 500, suggesting that all eigenmodes are physically active on cosmologically observable timescales.
5. **The WKB eigenvalue formula (4.6)** is parameter-free in the same way as the modified force law: k_n depends only on R (and n, π), making the spectrum a clean prediction of the framework.

Appendix A: Original Graviton Force Derivation (from GravitonForce.md)

Graviton Force in the TauTimp Framework

Polynomial probability-density solution of the d'Alembert equation matched to Newtonian gravity

Framework anchors. - Onduscular radius: $R = 4.2355 \times 10^{22}$ m. - Speed of light: $c = 2.99792458 \times 10^{8}\ \mathrm{m\,s^{-1}}$. - Reduced Planck constant: $\hbar = 1.054571817 \times 10^{-34}$ J s. - Newton constant: $G = 6.67430 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$.

A.1 Governing equation and ansatz

The graviton temporal-probability density $P(r,t)$ is postulated to satisfy the radial d'Alembert equation

$$\frac{1}{c^2}\frac{\partial^2 P}{\partial t^2} - \frac{\partial^2 P}{\partial r^2} = 0.$$

The graviton flux through a sphere of radius r at fixed t defines the modified gravitational force

$$F(r,t) = \frac{P(r,t)}{4\pi r^2}.$$

At $t = 0$ we look for a degree-2 polynomial ansatz

$$P(r,0) = a_0 + a_1 r + a_2 r^2,$$

subject to the cosmological boundary condition

$$P(2R,\,0) = 0.$$

Following the chosen double-root form (a tangential touch of P to zero at $r = 2R$):

$$\boxed{P(r,0) = A\left(1 - \frac{r}{2R}\right)^2.}$$

This automatically satisfies (A.1.4) and additionally enforces $\partial_r P(2R,0) = 0$, ensuring smooth tangency at the cosmological boundary.

A.2 Determination of the coefficients A.2.1 Newtonian-matching condition

For $r \ll R$ (terrestrial through galactic scales, $r < 10^{18}$ m), the bracket in (A.1.5) expands as

$$\left(1 - \frac{r}{2R}\right)^2 = 1 - \frac{r}{R} + \frac{r^2}{4R^2}.$$

The force (A.1.2) at $t = 0$ becomes

$$F(r,0) = \frac{A}{4\pi\, r^2}\left(1 - \frac{r}{R} + \frac{r^2}{4R^2}\right).$$

Equating with Newton's law $F_{\mathrm{N}}(r) = GMm/r^2$ in the small-r limit:

$$\boxed{A = 4\pi\, GMm.}$$

A.2.2 Explicit coefficients

$$P(r,0) = 4\pi GMm - \frac{4\pi GMm}{R}\, r + \frac{\pi GMm}{R^2}\, r^2,$$

$$\boxed{\begin{aligned} a_0 &= 4\pi\, GMm, \\ a_1 &= -\frac{4\pi\, GMm}{R} = -\frac{a_0}{R}, \\ a_2 &= \frac{\pi\, GMm}{R^2} = \frac{a_0}{4R^2}. \end{aligned}}$$

A.3 Modified force law and fractional deviation

$$\boxed{F(r,0) = \frac{GMm}{r^2}\left(1 - \frac{r}{2R}\right)^2.}$$

The fractional deviation is parameter-free: $\boxed{\delta(r) \equiv \frac{F(r,0) - F_{\mathrm{N}}(r)}{F_{\mathrm{N}}(r)} = -\frac{r}{R} + \frac{r^2}{4R^2}.}$

At $r = 10^{18}$ m, $\delta \approx -2.36 \times 10^{-5}$; deviations are below observational sensitivity for all Solar-System and galactic scales.

A.4 d'Alembert time evolution

$$\boxed{P(r,t) = \frac{A}{2}\left[(1 - |r - ct|/2R)_+^2 + (1 - |r + ct|/2R)_+^2\right].}$$

A.5 Wave-function and Heisenberg analysis

$$\psi(r,0) = \sqrt{A_{\mathrm{norm}}}\,(1 - r/(2R)), \quad A_{\mathrm{norm}} = 3/(2R).$$

Moments: $\langle r\rangle = R/2, \quad \Delta r = R\sqrt{3/20}, \quad \Delta p = \sqrt{3}\,\hbar/(2R), \quad \Delta r \cdot \Delta p = \frac{3\sqrt{5}}{20}\,\hbar.$

The product $\Delta r \cdot \Delta p \approx 0.6708 \cdot \hbar/2$ lies *below* the standard Heisenberg bound, which signals that $\psi(r,0)$ is a **field amplitude** (with $\psi(0) \neq 0$ breaking Hermiticity of $\hat{p}$) rather than a bound-state wave function.

A.6 Cross-link to the present derivation

The double-root polynomial $(1 - r/(2R))^2$ of the original force derivation and the Airy-function eigenmodes of the present spherical-PDE derivation describe **two complementary regimes** of the same TauTimp framework:

- **Static / d'Alembert regime (Appendix A):** flat 1D d'Alembert equation in the radial variable; gives the modified Newton force law via direct polynomial matching.
- **Dynamical / variable-coefficient regime (Sections 1–7):** full spherical wave equation with R/r coefficient; admits an Airy-spectrum of normal modes whose ground period is $\sim 18\ Myr$.

Both yield parameter-free predictions depending only on R, providing two independent consistency checks of the onduscular radius value $R = 4.2355 \times 10^{22}$ m.